18: Tips for Case Studies

Stat 230 - Spring 2026

```r
library(tidyverse)
library(ggformula)
library(ggResidpanel)
library(bayesrules)
library(patchwork)
```

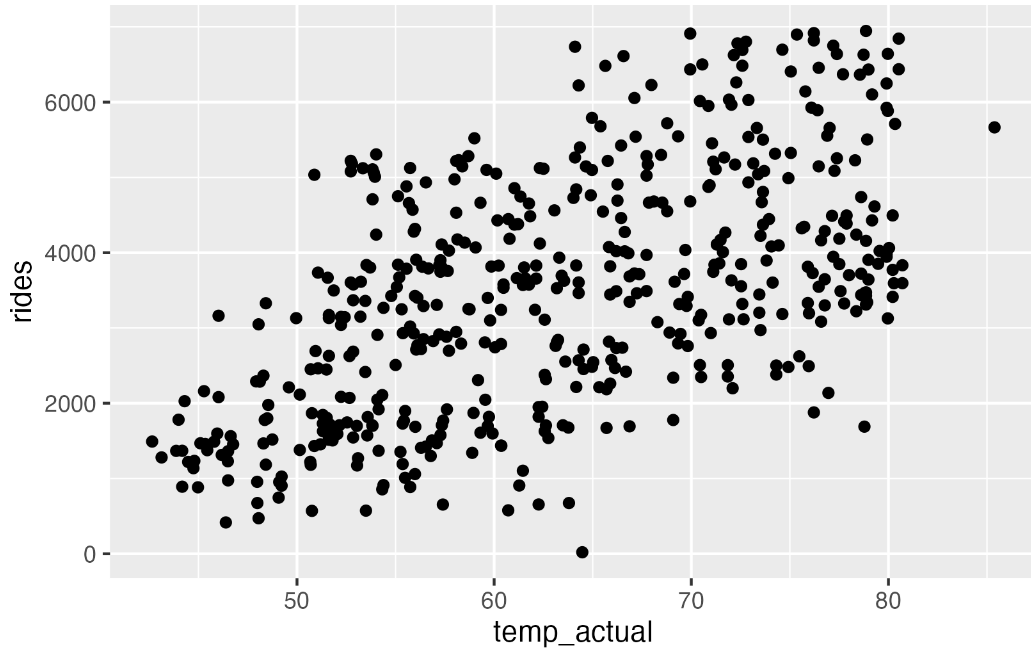
```r
mod <- lm(rides ~ temp_actual, data = bikes)
```

1 Figure size

In the first few lines of your code chunks, you can specify figure output.

Default:

```r
gf_point(rides ~ temp_actual, data = bikes)
```

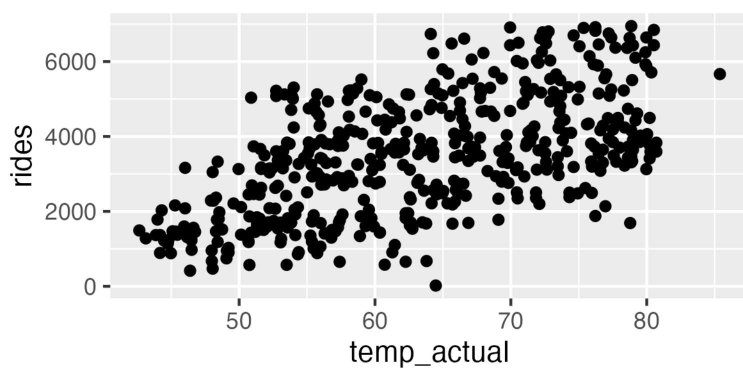


with

```
#| fig-width: 4
#| fig-height: 2
```

at top of chunk:

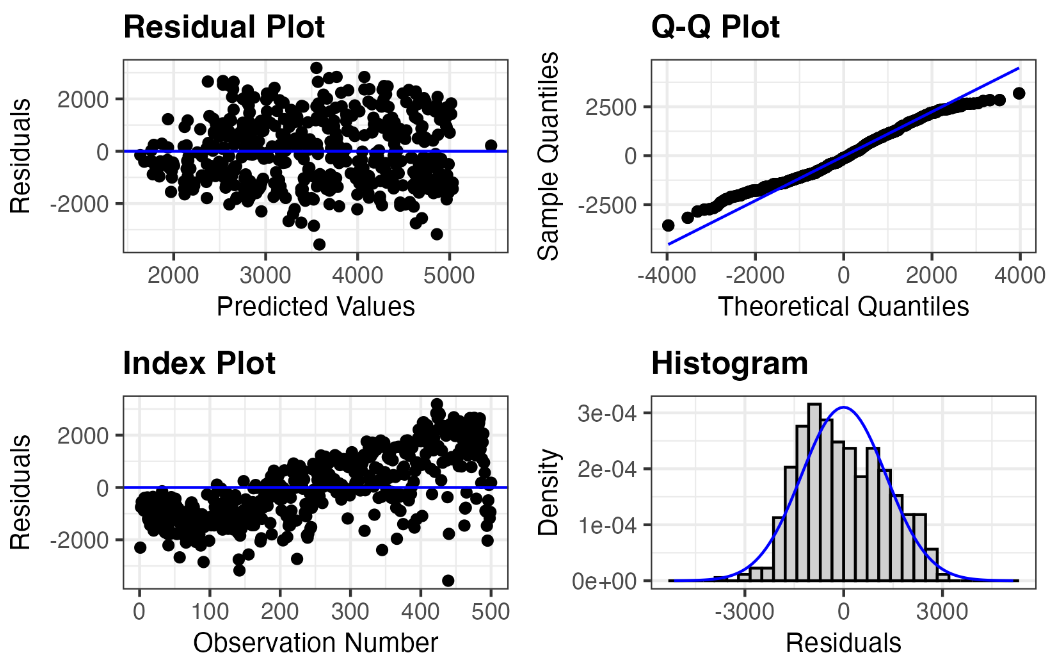
```
gf_point(rides ~ temp_actual, data = bikes)
```



2 Choosing a couple of residual plots

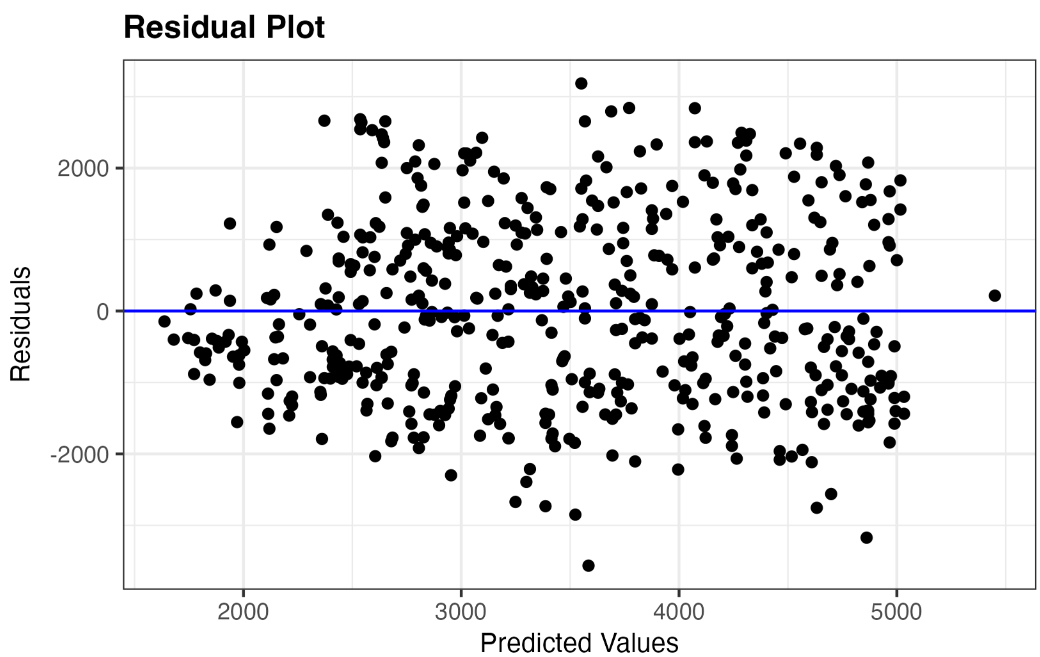
Sometimes, you might want to include some residual plots instead of the full panel (For example, you want to mention a weird trend in the index plot but all other residual plots are uninteresting). Specify using `plots` argument.

```r
resid_panel(mod)
```



vs

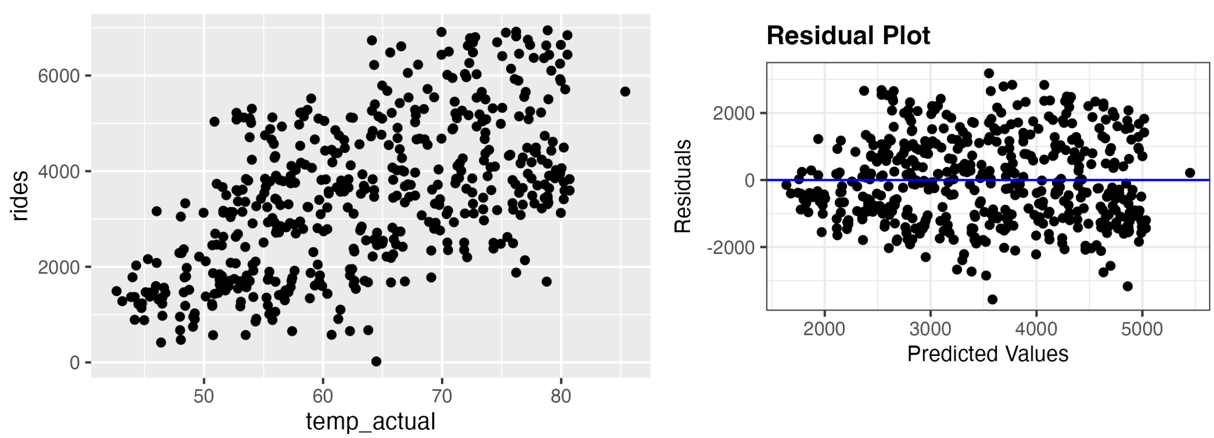
```r
resid_panel(mod, plots = "resid")
```



3 Combining graphics

Use the {patchwork} package to save individual plots and put them side by side

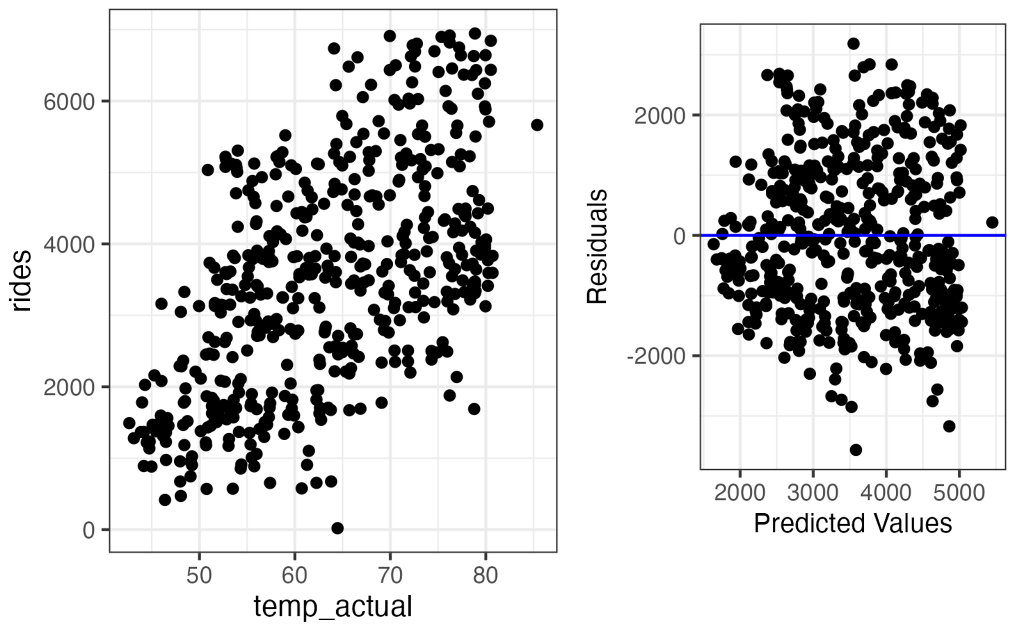
```
p1 <- gf_point(rides ~ temp_actual, data = bikes)
p2 <- resid_panel(mod, plots = "resid")

p1 + p2
```



4 Customize theme

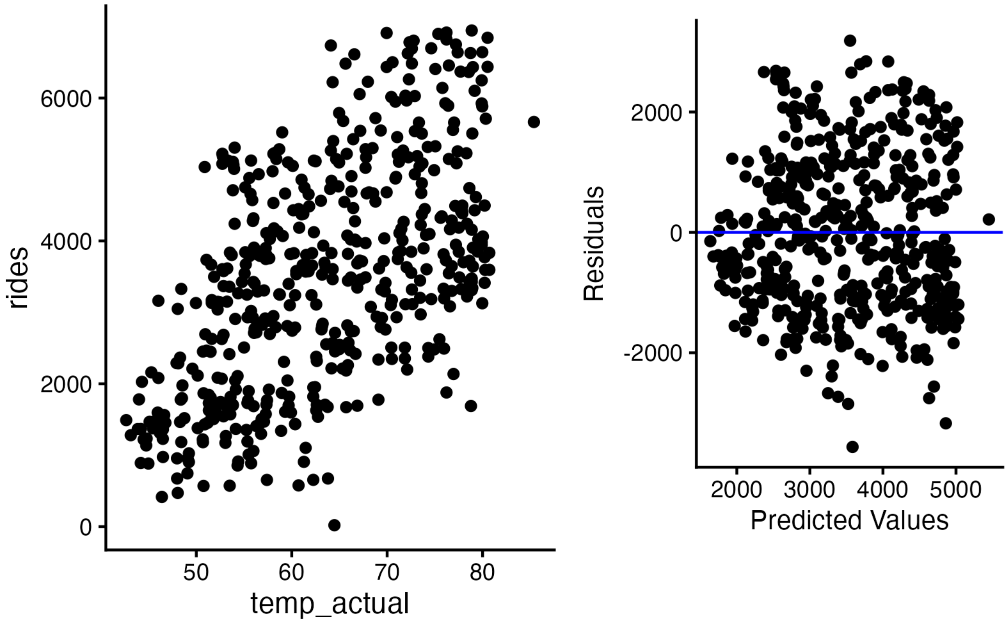
Oh no! The plots above look terrible together because they use different themes. *Note:* setting a theme is different with "regular plots" compared to resid panels

```
p1 <- gf_point(rides ~ temp_actual, data = bikes) + theme_bw()
p2 <- resid_panel(mod, plots = "resid", theme = "bw", title.opt = FALSE)

p1 + p2
```



Classic theme:

```
p1 <- gf_point(rides ~ temp_actual, data = bikes) + theme_classic()
p2 <- resid_panel(mod, plots = "resid", theme = "classic", title.opt = FALSE)

p1 + p2
```



Note: I know they show up as different sizes. I am not aware of a workaround besides using `augment()` and making your own residual plots.

5 Tables

The default summary output takes up too much space in a case study report:

```
summary(mod)
```

```
Call:
lm(formula = rides ~ temp_actual, data = bikes)

Residuals:
    Min      1Q  Median      3Q     Max 
-3564.3 -1009.3  -116.4   959.1  3184.6 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -2168.77     363.64  -5.964 4.67e-09 ***
temp_actual    89.23       5.67  15.739  < 2e-16 ***
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1288 on 498 degrees of freedom
Multiple R-squared:  0.3322,	Adjusted R-squared:  0.3308
F-statistic: 247.7 on 1 and 498 DF,  p-value: < 2.2e-16
```

Instead, use `broom::tidy()` for a more compact output.

```r
broom::tidy(mod)
```

```
# A tibble: 2 x 5
  term        estimate std.error statistic  p.value
  <chr>          <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)  -2169.     364.       -5.96 4.67e- 9
2 temp_actual     89.2      5.67     15.7  1.35e-45
```

You can also specify confidence intervals:

```r
broom::tidy(mod, conf.int = TRUE)
```

```
# A tibble: 2 x 7
  term        estimate std.error statistic  p.value conf.low conf.high
  <chr>          <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
1 (Intercept)  -2169.     364.       -5.96 4.67e- 9  -2883.    -1454.
2 temp_actual     89.2      5.67     15.7  1.35e-45     78.1      100.
```

And to make it look EVEN nicer, pass it into `knitr::kable()`:

```r
broom::tidy(mod, conf.int = TRUE) |>
  knitr::kable()
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-2168.77362	363.641906	-5.964036	0	-2883.2350	-1454.3122
temp_actual	89.23389	5.669703	15.738723	0	78.0944	100.3734

which has a `digits` option:

```r
broom::tidy(mod, conf.int = TRUE) |>
  knitr::kable(digits = 2)
```

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-2168.77	363.64	-5.96	0	-2883.24	-1454.31
temp_actual	89.23	5.67	15.74	0	78.09	100.37

6 Equations

If you are doing your report in .qmd, you might want to include some math! To include an inline equation, use single $: $y = \beta_0 + \beta_1 x_1 + \varepsilon, \varepsilon \sim N(0, \sigma^2)$.

To give an equation its own line, use double $$

$$y = \beta_0 + \beta_1 x_1 + \varepsilon, \varepsilon \sim N(0, \sigma^2)$$

(this typsetting language is called LaTeX and there are a bunch of resources available online)

- https://latexeditor.lagrida.com/
- https://webdemo.myscript.com/views/math/index.html#